STAC AI BOOTCAMP NYC



From language to credible predictions and decisions

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Why this section: can the result survive deployment?

Notebook: `05_alpha_risk.ipynb`

Decision: whether an apparent improvement merits a forward trial.

- Section 4: does adaptation improve the task on held-out cases?
- Section 5: could this workflow have made these decisions at the time?
- A fluent explanation, a correct extraction and a profitable action are different achievements.

Roadmap: from information to a credible decision

Each stage resolves a different reason a backtest can mislead.

Stage and time	Question answered
1. Define the task · 7 min	What is predicted, when, and for which action?
2. Reconstruct the past · 10 min	Which evidence, labels and model were available?
3. Test the complete workflow · 10 min	Does it beat a fair baseline after costs?
4. Diagnose and replay · 13 min	What caused the result, and can we reproduce it?
5. Decide what to claim · 5 min	What remains unproven before a forward trial?

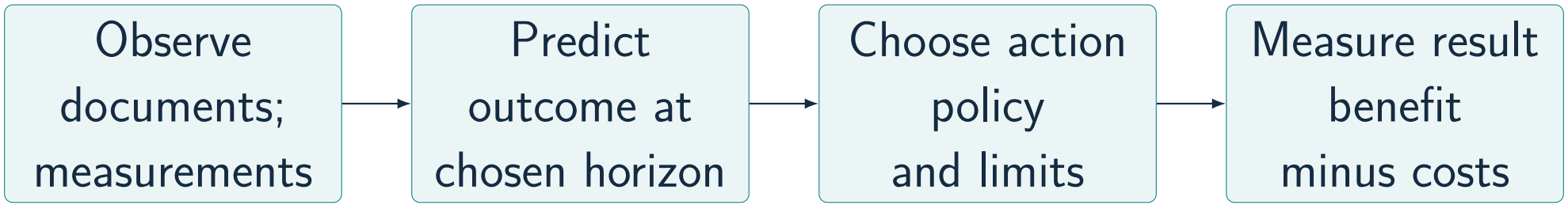
Five outcomes for this session

Use these as the acceptance checklist for a proposed experiment.

- Specify the target, horizon, information boundary and action.
- Construct a chronological replay with valid source and label availability.
- Compare against baselines without tuning on the final test.
- Separate predictive quality from decision value and costs.
- Locate a failure in evidence, interpretation, policy or execution.

Separate observation, prediction and action

Decision: which component is responsible for the claimed improvement.



A language model may extract evidence, produce a feature, or propose an action. Specify which role is being tested.

Define the prediction before choosing the model

$$\widehat{y}_{t,h} = f_{\theta_t}(x_t), \qquad y_{t,h} = \text{outcome over } (t, t+h].$$

- t: decision time; h: forecast horizon, in the same time unit.
- x_t: inputs available at t; f: prediction function; θ_t: its fitted parameters available then.
- $\widehat{y}_{t,h}$: predicted outcome; $y_{t,h}$: observed outcome. A hat marks an estimate; $(t, t+h]$ excludes the start and includes the end.
- Example: use reports available at 09:00 to predict failure over seven days. Define the outcome and intervention costs first.

A point-in-time labelled training record

Field	Illustrative record
Decision	Fictional stock ABC, Monday 09:00.
Input available then	Archived release and market data available by 09:00.
Outcome label	1 if the next five-trading-day excess return is positive; 0 otherwise.
Label availability	Only after the outcome window and any data delay.

The future outcome is a training label, never part of the original forecast input.

Changing regimes: what must remain useful?

- A relation learned in one period may fail when liquidity, behavior or market structure changes.
- An LLM can propose news-based indicators or hypotheses; their usefulness still needs testing.
- Evaluate later periods and stressed regimes; define triggers for reassessment.

No strict stationarity is needed to write a forecast equation. Generalization still needs some stability or an effective, validated adaptation rule.

Turn language into a reproducible measurement

Decision: what exactly will be tested across time.

Component	Record it explicitly
Input	Source IDs, saved text, availability times and versions.
Transformation	Prompt, model version, retrieval settings and parser.
Feature or forecast	A defined quantity: delay duration, exposure, event probability.
Failure handling	Missing, contradictory or unparseable evidence leads to a declared fallback.

Chronological rows do not guarantee historical knowledge

$$\mathcal{I}_t = \{\text{records and versions actually available by } t\}, \qquad x_t = g(\mathcal{I}_t).$$

- t: decision time; $\mathcal{I}_t$: information set available then; braces denote a set.
- g: declared input-building procedure; x_t: resulting model input.
- Event time, availability time and revision time can differ. All clocks must share a specified time zone.
- Read: construct the input only from eligible historical versions. A date-sorted table alone does not prove eligibility.

A saved URL is not a saved historical document

Decision: which version may enter a historical replay.

Time	What the workflow could see
Monday 09:00	Report version 1: demand grew 2%.
Monday 09:05	Decision made using version 1.
Tuesday 12:00	Same URL updated: growth revised to 5%.
Replay today	Fetching the URL now returns information unavailable on Monday.

Keep: content snapshots, version identifiers, availability records and extraction history.

The model has a knowledge history too

Decision: retrospective analysis or a feasible historical strategy?

- Filtering retrieved documents does not remove knowledge encoded in model weights.
- A current model applied to an old event may recognize its later outcome.
- Use documented model vintages where possible; test prospectively on newly resolved events.
- If historical availability cannot be established, narrow the claim to retrospective extraction.

Question: which claim does the evidence justify?

Test objective: distinguish source timing from model timing.

- A 2026 model reads only documents dated before each 2018 decision.
- Its simulated strategy performs well during 2018.
- A. This proves the model could have traded profitably in 2018.
- B. This supports a retrospective result, with model-knowledge contamination unresolved.
- C. The document filter guarantees that no later information was used.

Answer: B — the model vintage remains unresolved

Decision: what further evidence is required before claiming foresight.

- The document filter restricts retrieval, not the contents of the weights.
- A reproducible historical extraction result can still be informative.
- A forward trial with frozen settings tests prediction under a defensible information boundary.

Evaluate as the forecast origin moves forward

Decision: whether the model is learning only from the past.



Freeze each prediction before observing its outcome. Evaluate the horizon actually required.

A past row may still have an unresolved label

$$\mathcal{D}_t = \{(x_i, y_i) : a(x_i) \le t,\ a(y_i) \le t\}.$$

- t: training cutoff; i: example index; x_i: input; y_i: resolved outcome label.
- $a(z)$: availability time of record z; $\mathcal{D}_t$: eligible training pairs. Colon means "such that; $\le$ means no later than.
- Read: both the input and its label must be available before fitting. Use one time zone and include reporting delays.
- A Monday input with a seven-day outcome is not a resolved example for Friday training.

Implement the split; do not mistake it for an audit

Decision: what the library handles and what you must supply.

Python building block

```
TimeSeriesSplit(n_splits=3, gap=5)
```

- Produces ordered train/test indices; a gap excludes rows near the boundary.
- Choose the gap from label overlap and availability, not an arbitrary convention.
- The splitter cannot authenticate document versions, model vintages or timestamps.
- For irregular events, use actual availability times and outcome intervals.

Fit every learned transformation inside the training window

Decision: where preprocessing and calibration belong.

Leakage route	Repair
Normalize using all observations	Estimate the scaler from the training window only.
Select features using final test outcomes	Select within training/validation; leave final test untouched.
Tune a prompt after inspecting test failures	Treat those cases as development; reserve another test.
Fit a probability mapping on test labels	Fit on separate available calibration data.

A convincing comparison holds the opportunity set fixed

Decision: whether the language model adds incremental value.

Comparator	What it establishes
No action / existing operating policy	Whether changing behavior is worthwhile at all.
Simple feature or deterministic rule	Whether complex language processing adds useful information.
Same model, different evidence	How evidence quality changes results.
Same evidence, different policy	How much is due to the action rule.

Searching many designs changes what the best score means

Decision: how much confidence to place in a selected winner.

- Prompts, models, horizons, filters and thresholds are all research choices.
- Selecting the best historical result can select noise.
- Log attempted designs; separate tuning periods from final evaluation.
- Repeat across later periods and examine the whole search process.

Language likelihood and event probability answer different questions

$$P(\text{answer tokens} \mid \text{prompt}) \neq P(Y_{t,h} = 1 \mid \mathcal{I}_t).$$

- P: probability; $\mid$: conditional on; $\neq$: not generally equal.
- t: decision time; h: horizon; $Y_{t,h}$: binary event indicator (1 = event, 0 = no event).
- $\mathcal{I}_t$: information available at t. Both probabilities are unitless, but describe different objects.
- Token likelihood concerns wording. Event probability requires a defined outcome and validation against resolved cases.

Evaluate probabilities and consequences separately

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2, \qquad y_i \in \{0, 1\}.$$

- N: number of evaluated cases; i: case index; Σ: sum over those cases.
- p_i: predicted event probability; y_i: observed binary outcome. $\in$ means belongs to; 1 = event and 0 = no event.
- Read: average squared probability error. Score and probabilities are unitless; lower is better for this loss.
- Use a common resolved target and paired cases. The score measures forecast quality, not intervention costs or trading value.

Choose a policy using explicit consequences

$$a^*(x) = \arg\min_{a \in \mathcal{A}} \mathbb{E}[L(a, Y) \mid x].$$

- x: available input; Y: uncertain outcome; a: action; $\mathcal{A}$: permitted action set.
- $L(a, Y)$: loss in declared cost units; $\mathbb{E}[\cdot \mid x]$: conditional average; arg min: action attaining the smallest value; a^*: chosen action.
- Inspection example: C = inspection cost; D = failure loss; p = event probability. If inspection prevents all loss, inspect when $pD > C$.
- Equivalently $p > C/D$ for $D > 0$. Costs share units; p and C/D are unitless. Validate probability and intervention assumptions.

For trading, measure the return after implementation

$$R_{t+1}^{\text{net}} = w_t r_{t+1} - c\,|w_t - w_{t-1}|.$$

- t: rebalance time; $t+1$: next return interval; $t-1$: previous rebalance.
- w_t, w_{t-1}: current and previous portfolio weights; r_{t+1}: asset return; R_{t+1}^{net}: net portfolio return (decimal units).
- c: proportional cost per unit weight traded; $|\cdot|$: absolute value. The weight change measures turnover.
- Read: exposure times return, less trading cost. This simplified model excludes funding, market impact and variable slippage unless added explicitly.

For risk, count severity and timing as well as errors

Decision: whether warnings improve risk management.

Measure	Why it changes the conclusion
Breach or event frequency	Too few warnings may miss problems; too many may be costly.
Loss severity	One large miss can dominate many small correct decisions.
Clustering and regime	Concentrated failures may expose an unusable conditional forecast.
Lead time and response cost	A correct warning may arrive too late or trigger wasteful intervention.

Question: which improvement is decision-relevant?

Test objective: distinguish prediction accuracy from policy value.

- Model A catches 8 of 10 failures and triggers 90 unnecessary inspections.
- Model B catches 7 of 10 and triggers 10 unnecessary inspections.
- A. A must be better because recall is higher.
- B. B must be better because it raises fewer false alarms.
- C. We need the costs, severity and effectiveness of intervention.

Answer: C — specify the loss and intervention effect

Decision: compare policies on the same outcomes and assumptions.

- Count what happens after each permitted action, including no action.
- Include false alarms, missed events, delays and intervention costs.
- Report predictive metrics separately so policy choices remain visible.

Diagnose the chain before changing the model

Decision: which intervention would address the observed failure.

Stage	Test that isolates the problem
Evidence	Replace a stale or missing source with a verified eligible source.
Interpretation	Hold evidence fixed; compare extracted facts with reviewed labels.
Policy	Hold the signal fixed; recompute the permitted action.
Execution	Compare authorized intent with actual dispatched action and cost.

Class Exercise: did the signal predict or read the future?

- Open `05_alpha_risk.ipynb`; run setup C01, then C02.
- Compare valid information with future-derived signals, gate off and on.
- C03: inspect signal availability against decision time for each row.
- C04: change fees; compare net returns with the zero-return no-trade baseline.

What to inspect before accepting the attractive result

Decision: whether the result supports prediction, enforcement or neither.

Look at	Interpretation
Signal availability versus decision time	Late information makes that simulated prediction infeasible.
Blocked order and its reason	The gate enforced the declared information rule.
Gross versus net result	Costs may remove the apparent economic benefit.
Same opportunity and return ledger	Differences can be attributed to the changed setting.

From a promising replay to a forward trial

Decision: what must be frozen before the next outcome arrives.

- Freeze the target, eligibility rules, model, prompt, policy and cost assumptions.
- Timestamp predictions and proposed actions before outcomes resolve.
- Keep abstentions, missing inputs and rejected actions in the evaluation.
- Predefine review triggers; examine recent performance and failure groups.

Return to the five learning objectives

What you should now be able to explain.

Objective	Evidence from this session
Define a testable target	Write unit, decision time, horizon and outcome rule.
Respect the information boundary	Check source versions, model vintage and label availability.
Design a fair comparison	Use rolling origins, frozen choices and explicit baselines.
Measure decision value	Separate probability quality, costs and consequences.
Diagnose a failed workflow	Trace evidence, interpretation, policy and execution.

Reading map: use each source for a specific decision

Methods are attributed; workshop examples are explicitly illustrative.

Source	Use in this lecture
Hyndman & Athanasopoulos, FPP3 ch. 5	Forecasting workflow, rolling origins, baselines and quantile evaluation.
scikit-learn official guidance	Temporal splitting, leakage prevention and calibration.
FRED/ALFRED real-time periods	Observation dates versus information vintages.
Bailey et al., backtest overfitting	Why selecting a winner from many historical trials needs scrutiny.